

machine learning algorithms for data mining. It packages tools for data pre-processing, classification, regression, clustering, association rules and visualisation. The various ways of accessing it are – Weka Knowledge Explorer, Experimenter, Knowledge Flow and a simple CL Explorer is a user-friendly graphical interface for two-dimensional visualisation of mined data. It lets you import the raw data from various file formats, and supports well known algorithms for different mining actions like filtering, clustering, classification and attribute selection. However, when dealing with large data sets, it is best to use a CL based approach as Explorer tries to load the whole data set into the main memory, causing performance issues. This software also provides a Java Appetiser for use in applications and can connect to databases using CJD. Weka has proved to be an ideal choice for educational and research purposes, as well as for rapid prototyping.

Rapid Miner

Rapid Miner is available in both FOSS and commercial editions and is a leading predictive analytic platform. Gartner, the US research and advisory firm, has recognised Rapid Miner and Knime as leaders in the magic quadrant for advanced analytic platforms in 2016. Rapid Miner is helping enterprises embed predictive analysis in their business processes with its user friendly, rich library of data science and machine learning algorithms through its all-in-one programming environments like Rapid Miner Studio. Besides the standard data mining features like data cleansing, filtering, clustering, etc, the software also features built-in templates, repeatable work flows, a professional visualisation environment, and seamless integration with languages like Python and R into work flows that aid in rapid prototyping. The tool is also compatible with weak scripts. Rapid Miner is used for business/commercial applications, research and education.

Orange

Python users playing around with data sciences might be familiar with Orange. It is a Python library that powers Python scripts with its rich compilation of mining and machine learning algorithms for data pre-processing, classification, modelling, regression, clustering and other miscellaneous functions. Orange also comes with a visual programming environment and its workbench consists of tools for importing data, and dragging and dropping widgets and links to connect different widgets for completing the workflow. The visual programming comes with an easy-to-use UI, with plenty of online tutorials for assistance. Due to the ease of programming and integration in Python, Orange can be a great take off point for novices and experts to plunge into data mining.

Knime

Knime is one of the leading open source analytic, integration and reporting platforms that comes as free

software and as well as a commercial version. Written in Java and built upon Eclipse, its access is through a GUI that provides options to create the data flow and conduct data pre-processing, collection, analysis, modelling and reporting. A Gartner survey reveals that customers are happy with the platform's flexibility, openness and smooth integration with other software like Weka and R. Given the small size of the company, Knime has a large user base and an active community. It makes use of Eclipse's extension mechanism capability to add plugins for the required functionalities like text and image mining. This software is ideal for enterprise use.

DataMelt

DataMelt or DMelt does much more than just data mining. It is a computational platform, offering statistics, numeric and symbolic computations, scientific visualisation, etc. To avoid digressing from our topic, I'll restrict myself to only covering its data mining capabilities. DMelt provides data mining features like linear regression, curve fitting, cluster analysis, neural networks, fuzzy algorithms, analytic calculations and interactive visualisations using 2D/3D plots and histograms. One can play around with its IDE (integrated development kit) or its functions can be called from applications using its Java API. Both community and commercial editions of DMelt are available on Linux, Mac OS, Windows and Android platforms. DMelt is a successor to the jHepWork and SCAVis programs, which some people working in data analysis might be familiar with. This software is well suited for students, engineers and scientists.

Apache Mahout

Mahout is primarily a library of machine learning algorithms that can help in clustering, classification and frequent pattern mining. It can be used in a distributed mode that helps easy integration with Hadoop. Mahout is currently being used by some of the giants in the tech industry like Adobe, AOL, Drupal and Twitter, and it has also made an impact in research and academics. It can be a great choice for anyone looking for easy integration with Hadoop and to mine huge volumes of data.

ELKI

ELKI is open source software written in Java and licensed under AGPLv3. This software focuses especially on cluster analysis and outlier detection with a compilation of numerous algorithms from both these domains. The software is accessed through a GUI that displays the results once the selected algorithm is run. ELKI's design goals are performance, scalability, completeness, extensibility and a modular design to welcome contributions. ELKI currently doesn't offer professional support and the software is optimised for use in science and research. Hence, this option works best for those in research.